

Notes: Shimer (JPE, 2005)

The Assignment of Workers to Jobs in an Economy with Coordination Frictions

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1 Big picture

- **Question.** How can a labor market with heterogeneous workers and heterogeneous jobs generate imperfect assortative matching, unemployment, vacancies, firm wage effects, and firm profit effects without relying on bargaining or exogenous mismatch?
- **Core mechanism.** The paper adds coordination frictions to a classic assignment model. Workers can observe posted wages and choose where to apply, but identical workers in a large anonymous market must use identical mixed strategies. Randomized applications mean that some firms get too many applicants while identical firms get none.
- **Why coordination frictions matter.** Because applications are stochastic, a high-productivity firm may fail to receive a high-productivity applicant and hire a lower-productivity worker, while a lower-productivity firm may receive and hire a higher-productivity worker. This creates endogenous mismatch.
- **Main decentralized result.** A competitive search equilibrium with type-contingent wage posting decentralizes the social planner's solution. Queue lengths are efficient and unique.
- **Wage implication.** Conditional on worker type, wages increase with firm productivity when worker and firm types are complements. High-productivity firms pay more because they are harder to get: higher wages compensate for a lower hiring probability.
- **Firm profit implication.** Conditional on firm type, profit increases with the productivity of the hired worker. Firms prefer to hire the most productive applicant, and this is consistent with decentralized profit maximization.
- **Matching implication.** The model predicts positive but imperfect assortative matching. More productive workers are relatively more likely to match with more productive firms, but the rank correlation is less than one because of coordination frictions.
- **Welfare implication.** Although unemployment and vacancies coexist, the allocation is constrained efficient under the anonymity restriction. The paper obtains a version of the first and second welfare theorems in a directed-search assignment model.
- **Economic takeaway.** Firm wage effects and imperfect sorting need not imply noncompetitive wage setting. They can arise in a competitive labor market when workers coordinate imperfectly over where to apply and firms can rank applicants.

2 Incremental contribution of the paper

- **From frictionless assignment to frictional assignment.** Classic assignment models imply perfect sorting and no coexistence of unemployment and vacancies. Shimer adds a minimal anonymity restriction that turns the assignment problem into a coordination problem.
- **Endogenous mismatch rather than assumed mismatch.** Random-search models often assume that workers must sample jobs randomly and therefore must accept mismatch. In this paper, workers can direct their search, and the planner could eliminate mismatch by segregating workers across job types. Mismatch occurs because it is optimal under coordination frictions.
- **Wage posting without exogenous bargaining.** Random-search models often require Nash bargaining or another wage-setting rule. Here, firms post type-contingent wages, workers choose application strategies, and wages emerge from competitive search equilibrium.
- **A link between firm fixed effects and competitive labor markets.** The model explains why a worker's wage can depend on employer quality after controlling for worker type. The explanation is selection into successful applications: econometricians observe the workers who got high-productivity jobs, not the workers who applied and failed.

- **Efficiency of directed search with heterogeneity.** The paper shows that decentralized wage posting can implement the social planner’s allocation even with heterogeneous workers and jobs, stochastic queues, unemployment, and vacancies.
- **Tractable closed form under no comparative advantage.** In the special case $x_{m,n} = h_m k_n$, workers follow threshold application rules. This generates a simple, transparent picture of mismatch and assortative matching.
- **Empirical predictions from theory.** The model produces testable comparative implications: wages rise with employer productivity conditional on worker type; firm profits rise with worker productivity conditional on firm type; and matched worker-firm types are positively, but not perfectly, correlated.

3 Data and measurement: model objects and observables

This is a theoretical paper rather than an empirical paper. The “data and measurement” objects are the model primitives and the observable implications that an econometrician could measure in matched worker-firm data.

3.1 Agents and types

There are M worker types and N firm/job types:

$$m = 1, \dots, M, \quad n = 1, \dots, N.$$

Let $\mu_m > 0$ be the measure of type m workers and $\nu_n > 0$ be the measure of type n firms. Each firm has one job, and each worker can fill at most one job.

Interpretation. Worker and firm heterogeneity is central. The paper is not about aggregate unemployment alone; it is about how heterogeneous workers are assigned to heterogeneous jobs when applications are not perfectly coordinated.

3.2 Production

A type m worker matched with a type n firm produces

$$x_{m,n} \geq 0,$$

where output is increasing in both m and n . Unemployed workers and vacant jobs produce zero. It is convenient to define

$$x_{0,n} = 0.$$

Interpretation. The production function governs both efficiency and sorting. Complementarity or supermodularity in $x_{m,n}$ determines whether better firms should be paired with better workers in a frictionless benchmark.

3.3 Application probabilities and queue lengths

The anonymity restriction implies that all type m workers use the same application probabilities. Let $p_{m,n}$ be the probability that a type m worker applies to a type n firm, with

$$\sum_{n=1}^N p_{m,n} = 1.$$

Because type m workers apply equally across the ν_n firms of type n , a type n firm receives an expected queue of

$$q_{m,n} = \frac{\mu_m p_{m,n}}{\nu_n}.$$

The expected queue of applicants with type at least m is

$$Q_{m,n} = \sum_{m'=m}^M q_{m',n}, \quad Q_{M+1,n} = 0.$$

Interpretation. Queue lengths are the key state variables. They translate decentralized application choices into hiring probabilities, unemployment probabilities, and vacancy probabilities.

3.4 Poisson application technology

The realized number of type m applications to a type n firm is Poisson with mean $q_{m,n}$. Thus a positive fraction of jobs receive no applications from a given type even when that type applies with positive probability.

Interpretation. This is the source of coordination friction. No worker or firm is confused, and no worker is searching blindly. The friction is that identical workers randomize independently, so some jobs get crowded and others are empty.

3.5 Observable empirical objects

The theory suggests several empirical objects that a matched worker-firm data set could measure:

- the worker's type or productivity proxy m ;
- the employer's type or productivity proxy n ;
- wages $w_{m,n}$ conditional on worker and firm type;
- profits $x_{m,n} - w_{m,n}$ conditional on firm and worker type;
- unemployment probabilities by worker type and application target;
- vacancies by firm type;
- the rank correlation between worker and firm types among realized matches.

Interpretation. The model is motivated by empirical facts such as firm wage effects, imperfect worker-firm sorting, and coexistence of unemployment and vacancies. It explains why these facts may arise in a competitive assignment economy.

4 Models and theoretical specifications

4.1 Social planner's objective

A planner tells workers where to apply and firms whom to hire. The planner treats identical agents anonymously. A type n firm employs a type m worker if it receives no applications from workers better than m and at least one application from type m . The probability of that event is

$$e^{-Q_{m+1,n}} (1 - e^{-q_{m,n}}).$$

Aggregate output is therefore

$$Y(q) = \sum_{n=1}^N \nu_n \sum_{m=1}^M e^{-Q_{m+1,n}} (1 - e^{-q_{m,n}}) x_{m,n}. \quad (1)$$

The feasibility constraint for each worker type is

$$\mu_m = \sum_{n=1}^N q_{m,n} \nu_n. \quad (2)$$

Interpretation. The planner chooses expected queues, not deterministic assignments. Because of anonymity, even the planner cannot tell identical workers to split perfectly across identical firms.

4.2 Planner first-order condition

Let v_m be the multiplier on the feasibility constraint. The planner chooses $q_{m,n} \geq 0$ to maximize output. The complementary slackness condition is

$$v_m \geq e^{-Q_{m,n}} x_{m,n} - \sum_{m'=1}^{m-1} e^{-Q_{m'+1,n}} (1 - e^{-q_{m',n}}) x_{m',n}, \quad q_{m,n} \geq 0, \quad (3)$$

with equality when $q_{m,n} > 0$.

Interpretation. The right-hand side is the marginal product of one extra type m application to a type n job. The application is valuable only when it changes whom the firm hires.

Proposition 1: efficient queues

Queue lengths $\{q_{m,n}\}$ and shadow values $\{v_m\}$ satisfying feasibility and complementary slackness are socially optimal. The social optimum is unique.

4.3 Competitive search equilibrium

In the decentralized economy, firms post type-contingent wages $w_{m,n}$. Workers observe wage offers and apply to one job. If a type m worker applies to a type n firm, the probability of being hired is

$$e^{-Q_{m+1,n}} \frac{1 - e^{-q_{m,n}}}{q_{m,n}}.$$

If type m workers apply to type n firms in equilibrium, they must be indifferent between that application and their best alternative:

$$v_m = e^{-Q_{m+1,n}} \frac{1 - e^{-q_{m,n}}}{q_{m,n}} w_{m,n}. \quad (4)$$

Equivalently,

$$q_{m,n} v_m = e^{-Q_{m+1,n}} (1 - e^{-q_{m,n}}) w_{m,n}. \quad (5)$$

Interpretation. A firm can attract a longer queue by offering a higher wage, but workers care about expected income, not just the posted wage. High wages may come with low hiring probabilities.

4.4 Firm profit maximization

A type n firm chooses a wage schedule to attract queues and maximize expected profit:

$$\sum_{m=1}^M e^{-Q_{m+1,n}} (1 - e^{-q_{m,n}}) (x_{m,n} - w_{m,n}). \quad (6)$$

After substituting workers' indifference constraints, the firm maximizes expected revenue from queues minus the cost of attracting those queues:

$$R_n(q_{1,n}, \dots, q_{M,n}) - \sum_{m=1}^M q_{m,n} v_m, \quad (7)$$

where

$$R_n(q_{1,n}, \dots, q_{M,n}) = \sum_{m=1}^M e^{-Q_{m+1,n}} (1 - e^{-q_{m,n}}) x_{m,n}. \quad (8)$$

Interpretation. This converts the decentralized problem into a standard profit maximization problem. Firms buy expected applications at prices v_m .

Proposition 2: decentralization and welfare

A competitive search equilibrium is described by queue lengths $\{q_{m,n}\}$, wages $\{w_{m,n}\}$, and expected incomes $\{v_m\}$ satisfying feasibility, complementary slackness, and the wage equation. The equilibrium is unique and has the same queues as the social optimum.

4.5 Equilibrium wage formula

When $q_{m,n} > 0$, the equilibrium wage is

$$w_{m,n} = \frac{q_{m,n} e^{-q_{m,n}}}{1 - e^{-q_{m,n}}} \left[x_{m,n} - \sum_{m'=1}^{m-1} e^{-(Q_{m'+1,n} - Q_{m,n})} (1 - e^{-q_{m',n}}) x_{m',n} \right]. \quad (9)$$

Interpretation. The wage equals the marginal value of a type m application conditional on hiring that worker. A worker may be very productive, but if another similar worker would also have applied, the marginal value of this particular application is lower.

4.6 Empirical prediction: wages rise with firm productivity

Suppose x is supermodular: worker and firm types are complements. Then:

Proposition 3: wages and firm quality

If x is supermodular, $Q_{m,n}$ is increasing in n when positive. More productive jobs are harder to get, are more likely to be filled, and pay higher wages conditional on worker type.

Interpretation. This explains why worker fixed-effect wage regressions may find employer effects. Workers in high-productivity firms earn more not because wage setting is noncompetitive, but because high-productivity jobs attract more competition and must pay compensating differentials for lower hiring probabilities.

4.7 Empirical prediction: profits rise with worker productivity

Proposition 4: profits and worker quality

Firm n 's profit $x_{m,n} - w_{m,n}$ is increasing in worker type m whenever $q_{m,n} > 0$. Thus firm profit is positively related to worker quality after conditioning on firm type.

Interpretation. Firms rank applicants by productivity and prefer the best applicant. The wage formula does not fully absorb the productivity difference because more productive workers may also be compensated through a higher probability of being hired.

4.8 Special case: no comparative advantage

The paper studies the tractable special case

$$x_{m,n} = h_m k_n, \quad (10)$$

where h_m and k_n are increasing. In this case, no worker type has a comparative advantage at any firm type. There is a threshold $\bar{k}_m$ for each worker type such that type m workers apply only to firms with $k_n \geq \bar{k}_m$. The threshold is increasing in the worker type. The queue length is

$$q_{m,n} = \min\{\log \bar{k}_{m+1}, \log k_n\} - \min\{\log \bar{k}_m, \log k_n\}. \quad (11)$$

The expected income of type m workers is

$$v_m = \sum_{m'=1}^m \bar{k}_{m'} (h_{m'} - h_{m'-1}). \quad (12)$$

Interpretation. Workers follow threshold application rules. More productive workers have higher thresholds, but even the most productive workers apply to a range of firms. This creates overlap in application sets and therefore mismatch.

4.9 Assortative matching

Let $e_{m,n}$ be the fraction of type m workers employed by type n firms. In the no-comparative-advantage case:

Proposition 6: log supermodularity and positive sorting

The employment matrix $e_{m,n}$ and the corresponding firm-side employment shares are log supermodular. Therefore, more productive workers are relatively more likely to be employed by more productive firms.

Interpretation. The model predicts positive assortative matching, but not perfect assortative matching. It reconciles a positive worker-firm rank correlation with observed mismatch.

5 Identification and theoretical design

Because this is a theory paper, the relevant “identification” is not econometric identification. It is the mapping from assumptions to mechanisms and observable predictions. The paper is careful to isolate which assumptions generate mismatch, wage effects, profits effects, and welfare results.

5.1 Frictionless benchmark

In the standard assignment model with transferable utility and no coordination frictions:

- if production is complementary, matching is perfectly positively assortative;
- identical workers earn the same wage regardless of employer;
- unemployment and vacancies do not coexist except at the bottom of the type distribution;
- rank correlation between worker and firm types is one in the ideal continuous case.

Interpretation. This benchmark is too sharp relative to the empirical evidence. The paper’s goal is to keep the assignment structure but weaken the unrealistic coordination assumption.

5.2 Anonymity as the source of coordination frictions

The paper imposes two anonymity restrictions:

1. firms can condition wages and hiring on worker type but not on a worker’s identity;
2. identical workers must use identical mixed application strategies in every subgame.

Interpretation. The point is not that workers are uninformed about jobs. They observe wages and choose where to apply. The friction is that identical workers cannot coordinate their identities to avoid applying to the same firm.

5.3 Why Poisson queues generate unemployment and vacancies

If a type m worker applies to type n firms with positive probability, each type n firm receives a random number of such applications. Some receive none and may remain vacant. Others receive multiple applicants and must reject some workers. Therefore unemployment and vacancies coexist even in a competitive equilibrium.

Interpretation. Unemployment is not caused by wage rigidity here. It is caused by congestion and coordination failure in application decisions.

5.4 Why high wages do not mean inefficient rents

High-productivity firms pay higher wages to compensate workers for lower hiring probabilities. Expected income is equalized across applications used by a worker type:

$$\text{expected income} = \text{hiring probability} \times \text{posted wage}.$$

Interpretation. This is the central empirical reinterpretation. Employer wage effects can be compensating differentials for job-finding risk in a competitive directed-search market.

5.5 Why the welfare theorem matters

The decentralized equilibrium replicates the planner’s queues. Therefore the model explains unemployment, vacancies, and mismatch without implying that decentralized markets are inefficient within the model’s constraint set.

Interpretation. This separates two issues: coordination frictions generate observable imperfections, but competitive wage posting can still allocate queues efficiently given those frictions.

5.6 Threats and limitations of the theoretical design

- **Single application.** Workers can apply to only one job. Multiple applications could reduce coordination frictions or change wage competition.
- **Static environment.** The model abstracts from repeated applications, on-the-job search, and waiting for better matches.
- **Type observability.** Firms can rank applicants by type. If firms could not rank applicants, low-productivity workers would crowd out high-productivity workers and the planner’s mismatch logic would change.
- **Anonymity restriction.** The model rules out identity-contingent coordination among identical workers. This is plausible in large anonymous markets but may be less plausible in small or networked labor markets.
- **No entry in the main model.** Firm entry is not endogenous, so the paper focuses on assignment and queues rather than vacancy creation decisions.

Interpretation. These limitations define the boundary of the contribution. The paper is most useful as a theory of assignment and wage dispersion in a large, anonymous, directed-search labor market.

6 Figures

6.1 Figure 1: Threshold function

Read.

- The horizontal axis is the worker type m and the vertical axis is the firm/job type n .
- The curve $\bar{k}_m$ is the threshold above which a type m worker applies.
- The shaded region is the set of job types that a type m worker applies to.
- The threshold is increasing in worker type, so better workers apply to better jobs more often.
- The shaded region remains wide even for high worker types, so even very productive workers apply across a range of firm types.

Economic message. The figure visualizes the paper’s main mechanism. More productive workers direct their search toward better firms, but the application sets overlap. This overlap creates mismatch: good firms may hire lower-type workers and lower firms may hire higher-type workers.

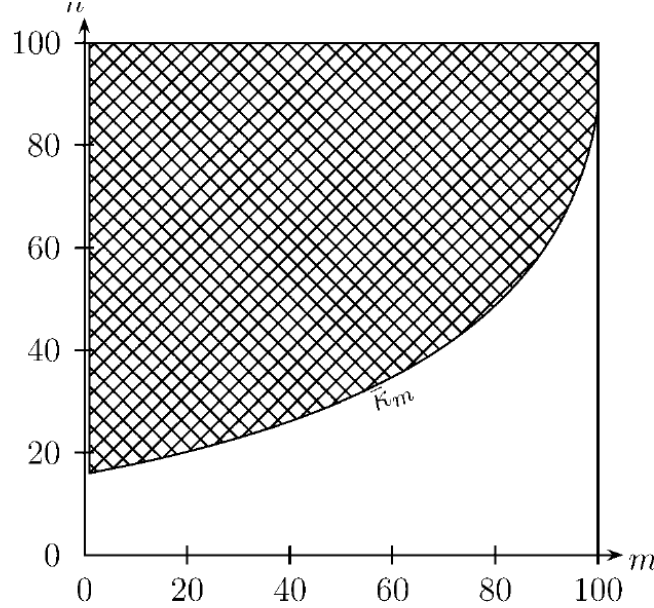


Figure 1: Threshold function. Original figure directly cropped from the paper. The shaded area indicates job types that each worker type applies to in the no-comparative-advantage example. The upward-sloping threshold shows positive sorting, while the wide shaded region shows imperfect sorting and endogenous mismatch.

Detailed interpretation. The curve has two messages. First, sorting is not random: high-productivity workers have higher thresholds and thus apply disproportionately to high-productivity jobs. Second, sorting is not perfect: workers of different types have overlapping application sets. The overlap is not a mistake or inefficiency in the planner’s problem. It helps insure high-productivity jobs against receiving no high-productivity applicants. In equilibrium, lower-productivity workers sometimes serve as a backup applicant pool for high-productivity firms.

How it fits the paper. Figure 1 supports Proposition 5 and the later assortative matching result. It shows why the model can produce both a positive worker-firm type correlation and visible mismatch. This is exactly the combination missing from frictionless assignment models.

6.2 Result Box 1: Social planner queues and decentralized equilibrium

Read

The planner chooses queue lengths $q_{m,n}$ to maximize output subject to feasibility. Firms in competitive search choose the same queues by posting wages that attract applicants.

Economic message

The competitive search equilibrium internalizes the queue externality through wage posting. Workers trade off posted wages against hiring probabilities, and firms pay wages equal to the marginal value of applications.

Detailed interpretation. This result is conceptually important because the model contains unemployment and vacancies but does not rely on a wedge between private and social incentives. The constrained planner and decentralized equilibrium agree on queues. Therefore the coordination friction creates randomness and mismatch, but it does not by itself create an inefficient queue allocation.

6.3 Result Box 2: Wage effects and firm effects

Read

Under supermodularity, wages are increasing in firm productivity conditional on worker type. A worker matched to a better firm earns more, even if the worker's type is held fixed.

Economic message

Firm wage effects can arise in a competitive model. The wage premium at better firms compensates workers for lower hiring probabilities at more desirable jobs.

Detailed interpretation. This result changes how one reads employer fixed effects. A high firm effect need not mean rents, efficiency wages, bargaining power, or monopsony. In Shimer's model, it can arise from selection into difficult-to-get jobs. The econometrician observes only successful matches, not the unsuccessful applications that make those high wages necessary.

6.4 Result Box 3: Profit effects and worker quality

Read

Firm profit is increasing in worker productivity conditional on firm type. The wage does not fully extract the higher output of better workers.

Economic message

Even though firms post wages competitively, they still prefer more productive applicants. This justifies the assumption that firms rank applicants by productivity.

Detailed interpretation. The profit result is useful because it connects the model to matched employer-employee evidence. A firm with higher-quality workers should earn higher profits, even after conditioning on firm characteristics. The reason is that wages reflect marginal application value, not the full match surplus in every state.

6.5 Result Box 4: Positive but imperfect assortative matching

Read

In the no-comparative-advantage case, the employment matrix is log supermodular: better workers are relatively more likely to be matched with better firms. But the correlation is less than one because of overlapping application sets.

Economic message

The model reconciles two facts: there is sorting in labor markets, but sorting is far from perfect. Coordination frictions produce the imperfection.

Detailed interpretation. Perfect sorting would eliminate many of the questions the paper wants to answer. If every worker type always matched with a unique firm type, it would be impossible to study how wages vary with employer quality holding worker type fixed or how profits vary with worker quality holding firm type fixed. Imperfect sorting creates the empirical variation that makes such questions meaningful.

7 Short summary

- The paper studies how heterogeneous workers are assigned to heterogeneous jobs in a large anonymous labor market.
- Workers observe wages and direct their search, but identical workers must use identical mixed strategies.
- Randomized applications generate coordination frictions: some jobs get too many applicants and some get none.
- A social planner chooses expected queues subject to anonymity and feasibility constraints.
- A competitive search equilibrium with type-contingent wage posting decentralizes the planner's allocation.
- Wages rise with firm productivity conditional on worker type when production is supermodular.
- Firm profits rise with worker productivity conditional on firm type.
- In the no-comparative-advantage case, workers use threshold application rules.
- Threshold rules imply overlap in application sets and therefore endogenous mismatch.
- The model predicts positive but imperfect assortative matching.
- The model explains firm wage effects, unemployment, vacancies, and mismatch without imposing exogenous random search or bargaining.
- Its main limitation is that it is static and assumes one application per worker.

8 Conclusion

8.1 Core conclusion

Shimer builds an assignment model in which coordination frictions arise naturally from anonymity in a large labor market. The model explains why unemployment and vacancies coexist, why workers are imperfectly matched to firms, why employer quality affects wages after conditioning on worker type, and why firm profits depend on worker quality after conditioning on firm type. The decentralized competitive search equilibrium implements the social planner's allocation, so these phenomena arise without assuming wage rigidity or inefficient bargaining.

8.2 Economic intuition

The intuition is that workers face a portfolio problem over applications. High-productivity firms pay more, but those jobs are harder to obtain because they attract more applicants. Workers therefore mix across a range of firms. Firms also want a mix of applicants: high-productivity applicants are desirable, but lower-productivity applicants provide insurance against the risk that no high-productivity worker applies. This overlap in application strategies creates mismatch.

8.3 Incremental takeaway

The paper's incremental contribution is to show that a small and plausible market-design restriction - anonymity among identical workers - can overturn the stark predictions of frictionless assignment while preserving tractability and efficiency. Instead of assuming workers randomly encounter jobs, the paper lets workers direct search toward wage offers. Instead of assuming mismatch, the paper derives it. Instead of imposing a wage-bargaining rule, the paper derives wages from competitive posting.

8.4 Relation to empirical work

The model gives a competitive interpretation of employer wage effects. In matched worker-firm regressions, a worker who moves to a higher-quality employer may receive a higher wage because the employer's job was harder to get, not necessarily because the employer shares rents. Similarly, firms with better workers may earn higher profits because wages do not fully absorb worker productivity. These predictions connect the theory to matched employer-employee data.

8.5 Limitations

The model is static, workers apply to only one job, and firm entry is not central. A dynamic extension would allow workers to reapply, firms to wait, and jobs to be geographically or temporally dispersed. The author notes that such an extension is necessary for quantitative applications and may strengthen assortative matching because high-productivity firms could wait for better applicants rather than hiring very low-productivity workers.

8.6 Bottom line

The central message is that labor-market mismatch does not require workers to be uninformed or wages to be noncompetitive. Even in a directed-search market with rational workers and wage-posting firms, anonymity and mixed application strategies generate coordination frictions. These frictions create unemployment, vacancies, imperfect sorting, employer wage effects, and profit-worker-quality correlations.